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Online Continual Learning on Class Incremental Blurry Task Configuration with Anytime Inference

ICLR 2022 accepted paper

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Related Works

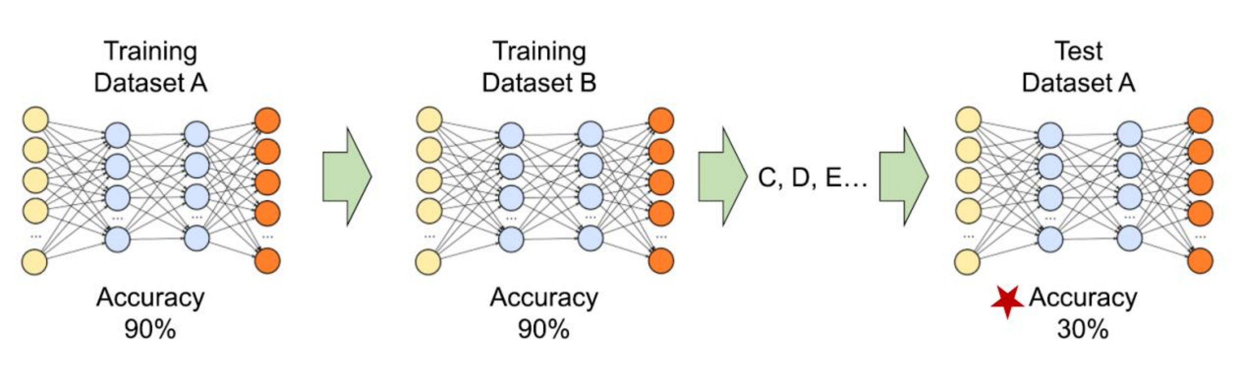
1. Continual Learning

= Incremental Learning
= Life-long Learning

*Concept to learn a model for **a large number of tasks sequentially without forgetting knowledge** obtained from the preceding tasks, where the data in the old tasks are not available anymore during training new ones.*

Catastrophic Forgetting

: 새로운 task에 대해 학습하게 되면 신경망모델이 이전에 배운 task에 대해서는 네트워크가 까먹는 현상으로 Transfer Learning에서 대표적으로 자주 발생함.



http://dmqm.korea.ac.kr/activity/seminar/266

Related Works

1. Continual Learning

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*Concept to learn a model for **a large number of tasks sequentially without forgetting knowledge** obtained from the preceding tasks, where the data in the old tasks are not available anymore during training new ones.*

Continual Enlargement of Data

현실에서는 연구의 방향이나 시장의 수요에 따라 데이터/클래스가 세분화 혹은 계속적으로 증가하고, 이에 따라 새로운 task도 추가됨
→ But, building individual model for each newly added task is too expensive!!

CL Settings

1. **전체 데이터로 재학습 불가능:** 학습에 사용된 subset데이터는 저장이 불가능하여 더 이상 사용할 수 없음. 사용하더라도 최소한으로 사용할 수 있는 방법을 고안하고자 함.
2. 여러 task를 하나의 모델에 **순차적으로 학습**하여 **최종적으로 모든 task의 수행이 가능**한 모델을 학습

Related Works

2. General CL Approaches

1) Regularization
: 이전 weight를 기억할 수 있도록 bias를 걸어주는 등의 regularization을 걸어줌

2) Parameter Isolation
: 각 task마다 activate할 parameter를 바꿔주어 과거에 학습된 parameter도 잊혀지지 않도록 함

3) Replay (=Rehearsal)
: fixed-sized buffer에 과거 데이터 일부를 저장하여 활용함

Algorithm 1 iCaRL CLASSIFY

input x // image to be classified
require $\mathcal{P} = (P_1, \dots, P_t)$ // class exemplar sets
require $\varphi : \mathcal{X} \to \mathbb{R}^d$ // feature map
  for $y = 1, \dots, t$ **do**
    $\mu_y \leftarrow \frac{1}{|P_y|} \sum_{p \in P_y} \varphi(p)$ // mean-of-exemplars
  end for
  $y^* \leftarrow \operatorname*{argmin}_{y=1,\dots,t} \|\varphi(x) - \mu_y\|$ // nearest prototype
output class label y^*

Algorithm 2 iCaRL INCREMENTALTRAIN

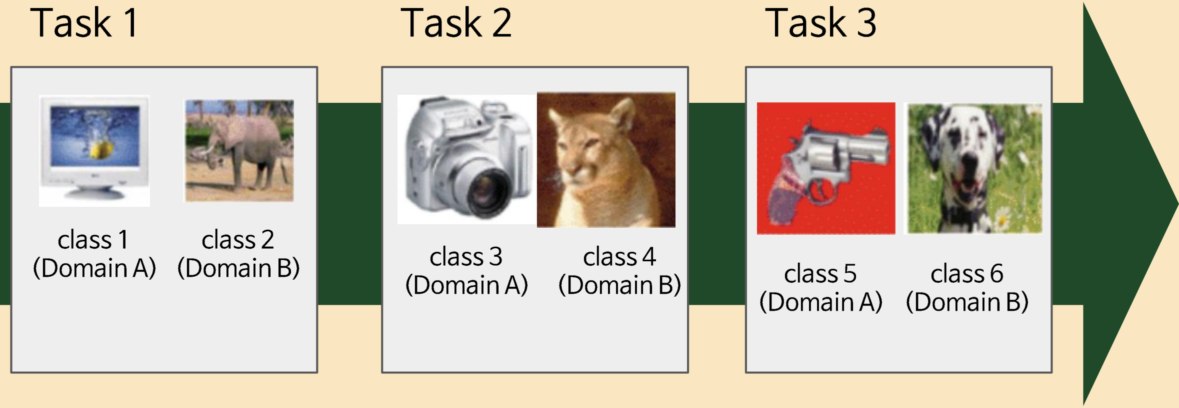
input $X^s, \dots, X^t$ // training examples in per-class sets
input K // memory size
require Θ // current model parameters
require $\mathcal{P} = (P_1, \dots, P_{s-1})$ // current exemplar sets
  $\Theta \leftarrow$ UPDATEREPRESENTATION$(X^s, \dots, X^t; \mathcal{P}, \Theta)$
  $m \leftarrow K/t$ // number of exemplars per class
  for $y = 1, \dots, s-1$ **do**
    $P_y \leftarrow$ REDUCEEXEMPLARSET(P_y, m)
  end for
  for $y = s, \dots, t$ **do**
    $P_y \leftarrow$ CONSTRUCTEXEMPLARSET(X_y, m, Θ)
  end for
  $\mathcal{P} \leftarrow (P_1, \dots, P_t)$ // new exemplar sets

클래스마다 representation을 배우고 클래스별로 n개의 exampler를 저장 해놓고 지속적으로 학습에 활용

Overcoming Catastrophic Forgetting by Incremental Moment Matching

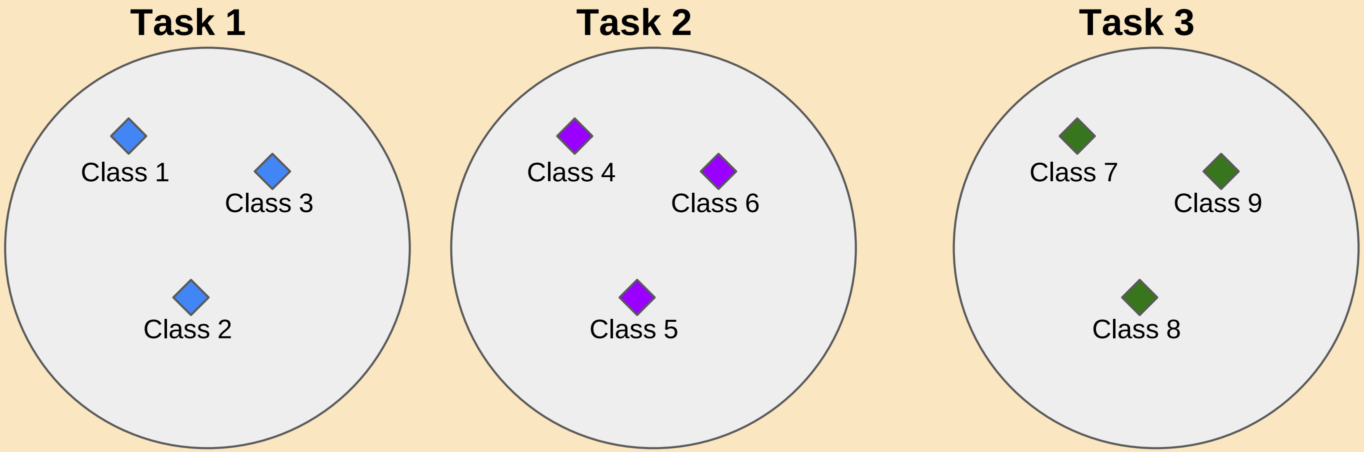
Related Works
3. CL situations

1. Task CL
2. Class CL
3. Domain CL



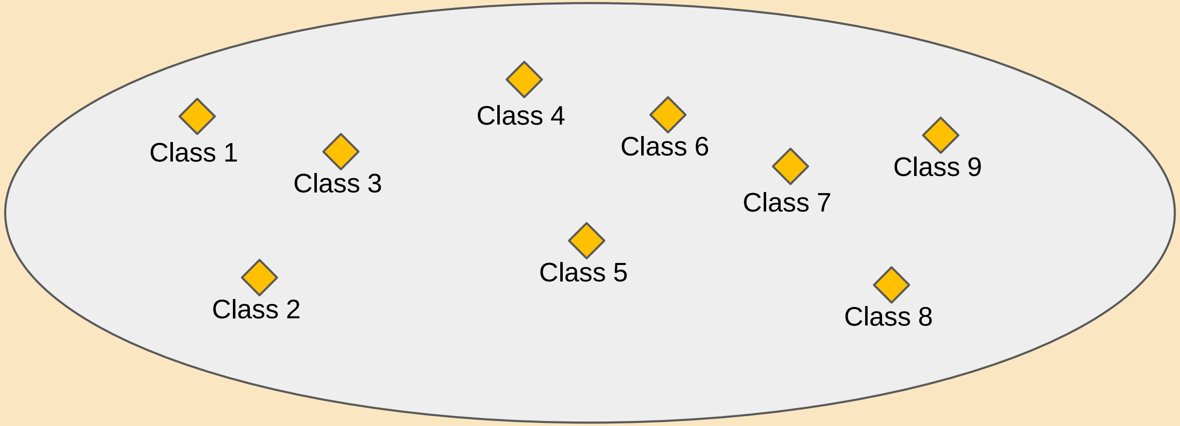
1. i-Blurry Task

✔ Disjoint Split



Thesis Problem
1. i-Blurry Task

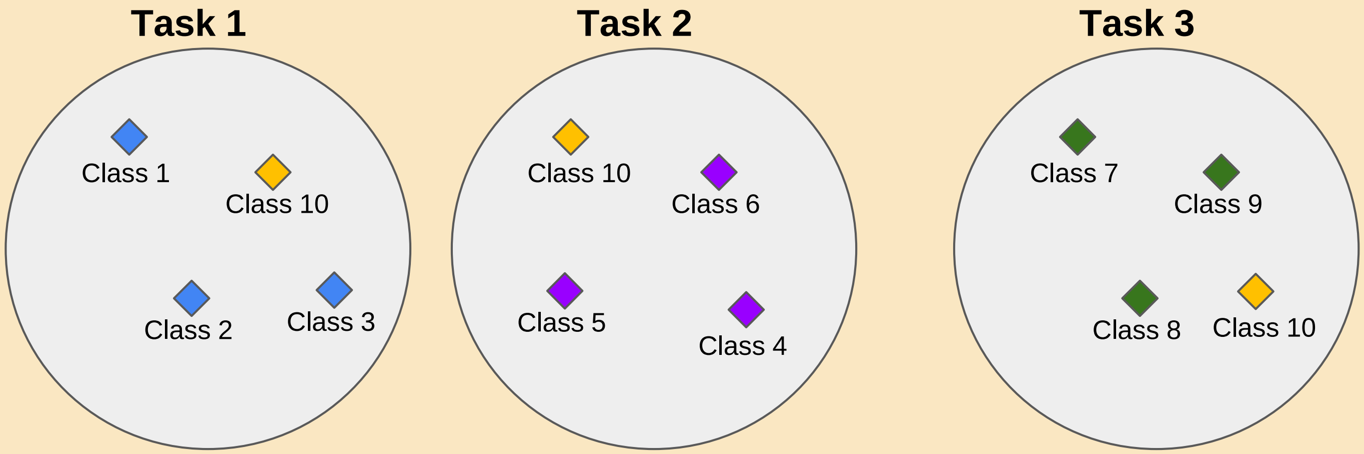
✔ **Blurry Split**



Thesis Problem

1. i-Blurry Task

✔ **i-Blurry-N-M**



1. i-Blurry Task

✔ i-Blurry-N-M

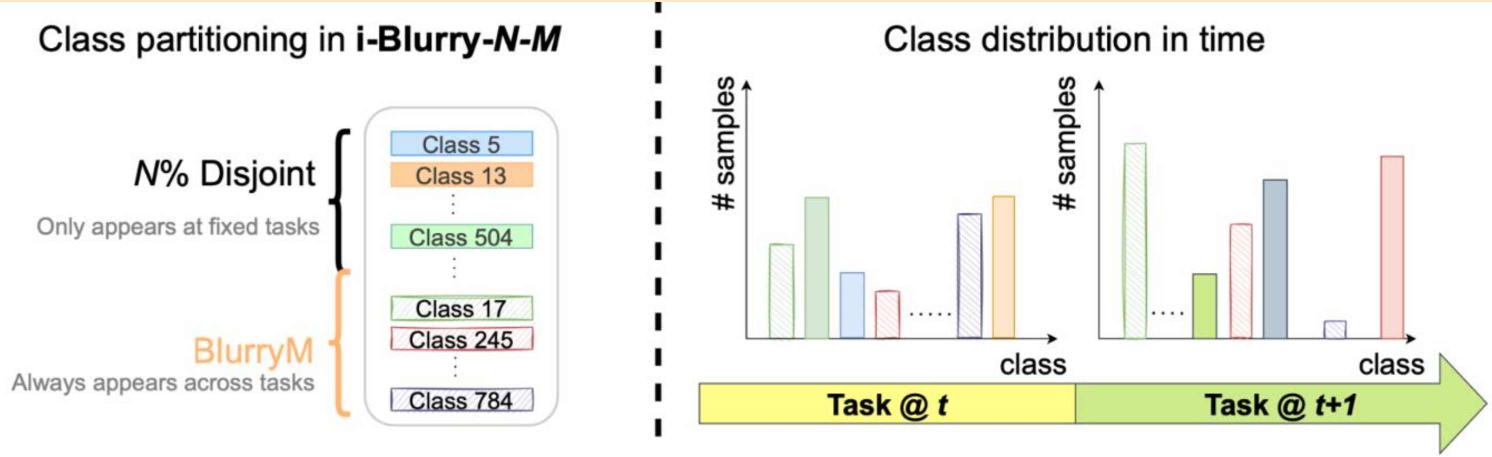
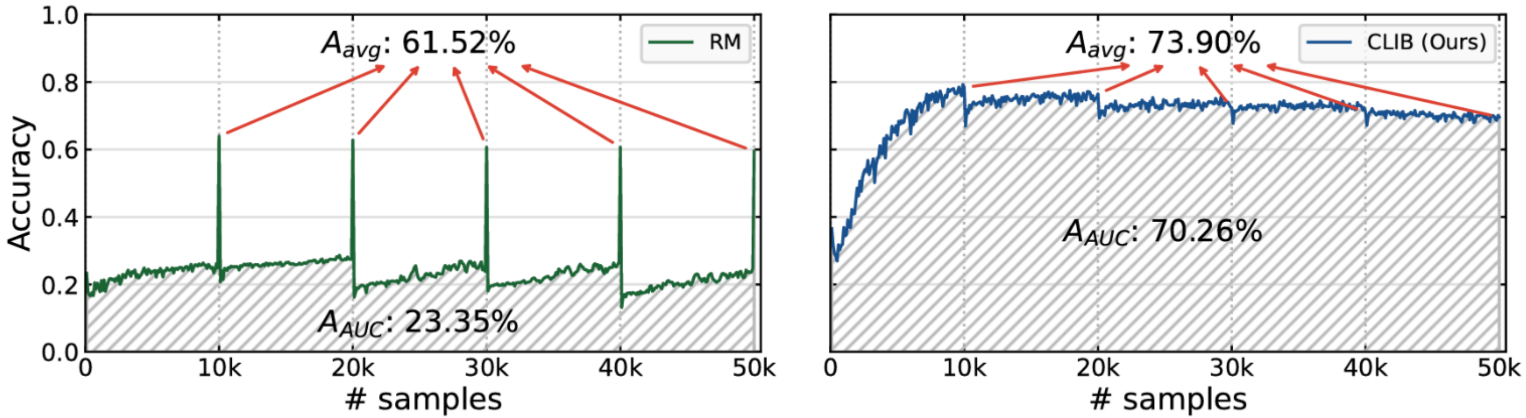


Figure 1: i-Blurry-N-M split. N% of classes are partitioned into the disjoint set and the rest into the Blurry$M$ set where M denotes the blurry level (Aljundi et al., 2019c). To form the i-Blurry-N-M task splits, we draw training samples from a uniform distribution from the 'disjoint' or the 'Blurry$M$' set (Aljundi et al., 2019c). The 'blurry' classes always appear over the tasks while disjoint classes gradually appear.

Thesis Problem

2. Metric for Practical Continual Learning



(a) Rainbow Memory (RM) (Bang et al., 2021) (b) CLIB

Figure 2: Comparison of A_{AUC} with A_{avg}. (a) online version of RM (Bang et al., 2021) (b) proposed CLIB. The two-stage method delays most of the training to the end of the task. The accuracy-to-{# of samples} plot shows that our method is more effective at any time inference than the two-stage method. The difference in A_{avg} for the two methods is much smaller than that of in A_{AUC}, implying that A_{AUC} captures the effectiveness at any-time inference better.

Q & A

Method

1. Sample-wise Importance Based Memory Management

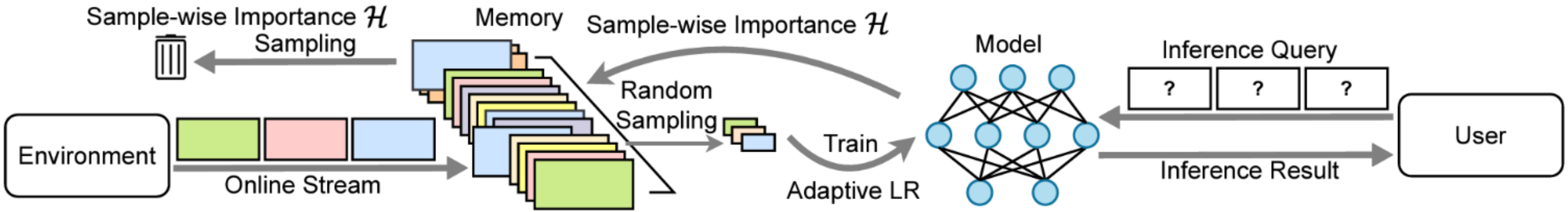
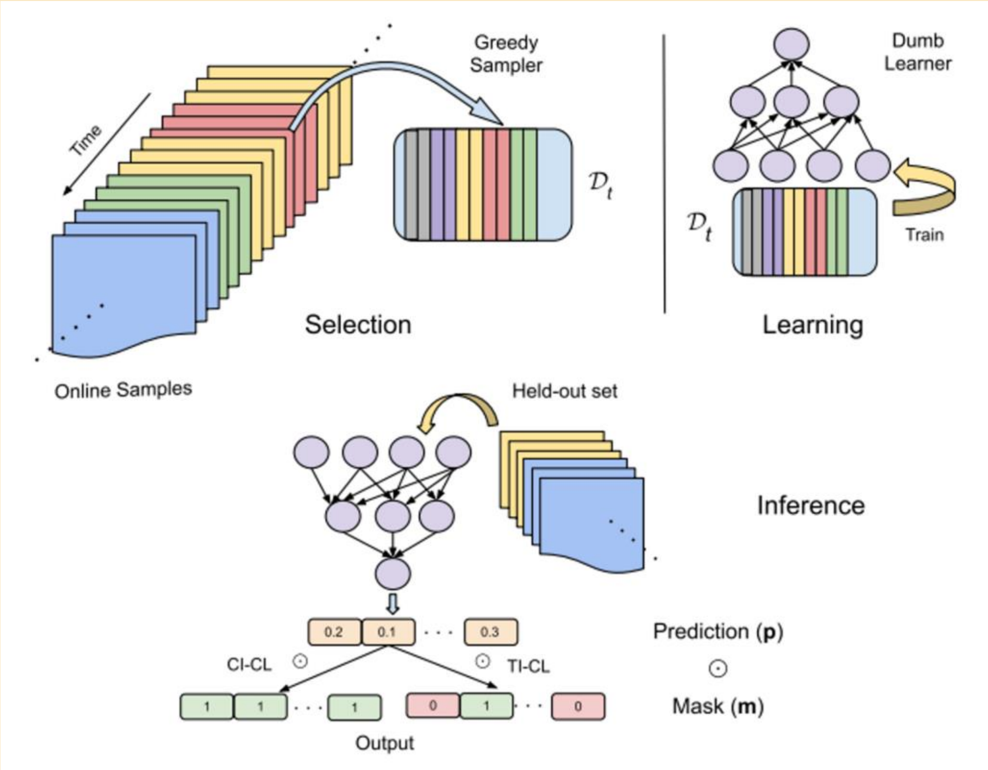


Figure 3: Overview of the proposed CLIB. We compute sample-wise importance during training to manage our memory. Note that we only draw training samples from the memory whereas ER based methods draw them from both the memory and the online stream.

1. samples that were **in the batch** when the **loss decreased**
2. samples that were n**ot in the batch** when the **loss increased**

high importance scores

2. Memory Only Training



Selection (Greed Sampler)

- 위 pseudocode와 같은 방식으로 입력 데이터를 클래스 별로 동일한 개수를 메모리에 저장

Learning (Dump Learner)

- 메모리에 저장된 데이터를 이용해서 학습

GDumb: A Simple Approach that Questions Our Progress in Continual Learning

2. Memory Only Training

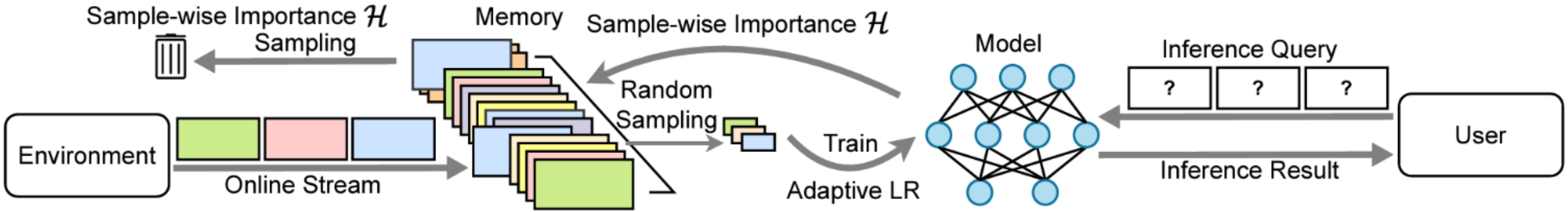


Figure 3: Overview of the proposed CLIB. We compute sample-wise importance during training to manage our memory. Note that we only draw training samples from the memory whereas ER based methods draw them from both the memory and the online stream.

Using the streamed samples directly will skew the **training to favor the recent samples more.**

- Propose to use samples only from the memory for training, without using the streamed samples
- The **memory works as a distribution stabilizer for streamed samples** through the memory update process and samples are used for training only with the memory.

Method

3. Adaptive Learning Rate Scheduler

Algorithm 3 Adaptive Learning Rate Scheduler

Input current LR η, current base LR $\bar{\eta}$, loss before applying current LR l_{before}, current loss l_{cur}, LR performance history $\mathcal{H}^{(\text{high})}$ and $\mathcal{H}^{(\text{low})}$, LR step $\gamma < 1$, history length m, significance level α

$l_{\text{diff}} = l_{\text{before}} - l_{\text{cur}}$ ▷ Obtain loss decrease

Update $l_{\text{before}} \leftarrow l_{\text{cur}}$

if $\eta > \bar{\eta}$ **then** ▷ If LR is higher than base LR

  Update $\mathcal{H}^{(\text{high})} \leftarrow \mathcal{H}^{(\text{high})} \cup \{l_{\text{diff}}\}$ ▷ Append loss decrease in high LR history

  if $|\mathcal{H}^{(\text{high})}| > m$ **then**

    Update $\mathcal{H}^{(\text{high})} \leftarrow \mathcal{H}^{(\text{high})} \setminus \{\mathcal{H}_1^{(\text{high})}\}$

  end if

else ▷ If LR is lower than base LR

  Update $\mathcal{H}^{(\text{low})} \leftarrow \mathcal{H}^{(\text{low})} \cup \{l_{\text{diff}}\}$ ▷ Append loss decrease in low LR history

  if $|\mathcal{H}^{(\text{low})}| > m$ **then**

    Update $\mathcal{H}^{(\text{low})} \leftarrow \mathcal{H}^{(\text{low})} \setminus \{\mathcal{H}_1^{(\text{low})}\}$

  end if

end if

if $|\mathcal{H}^{(\text{high})}| = m$ **and** $|\mathcal{H}^{(\text{low})}| = m$ **then** ▷ If both histories are full

  $p = \text{OneSidedStudentsTTest}\left(\mathcal{H}^{(\text{low})}, \mathcal{H}^{(\text{high})}\right)$

  ▷ Perform one-sided Student's t-test with alternative hypothesis $\mu_{\text{low}} > \mu_{\text{high}}$

  if $p < \alpha$ **then** ▷ If pvalue is significantly low

    Update $\bar{\eta} \leftarrow \gamma^2 \cdot \bar{\eta}$ ▷ Decrease base LR

    Update $\mathcal{H}^{(\text{low})}, \mathcal{H}^{(\text{high})} = \emptyset, \emptyset$ ▷ Reset histories

  else if $p > 1 - \alpha$ **then** ▷ If pvalue is significantly high

    Update $\bar{\eta} \leftarrow \frac{1}{\gamma^2} \cdot \bar{\eta}$ ▷ Increase base LR

    Update $\mathcal{H}^{(\text{low})}, \mathcal{H}^{(\text{high})} = \emptyset, \emptyset$

  end if

end if

if $\eta > \bar{\eta}$ **then** ▷ Alternately apply high and low LR (note that $\gamma < 1$)

  Update $\eta = \gamma \cdot \bar{\eta}$

else

  Update $\eta = \frac{1}{\gamma} \cdot \bar{\eta}$

end if

Output $\eta, \bar{\eta}, l_{\text{before}}, \mathcal{H}^{(\text{low})}, \mathcal{H}^{(\text{high})}$

It adaptively changes the LR in a **data-driven manner based** on how good the LR is for optimizing over the memory.

- ✔ The rate in which the LR can change is bounded and sudden changes in LR do not happen.
- ✔ It is important for CL methods because the data distribution changes over time and adapting to the current training data is more useful.

Q & A

Results

i-Blurry-50-10 Results

CLIB: Continual Learning for i-Blurry task

- Memory size
  - CIFAR10: 500, CIFAR100: 2000, TinyImageNet: 4000, ImageNet: 20000
- # of Task
  - CIFAR10, CIFAR100, TinyImageNet: 5 tasks, ImageNet: 10 tasks

Methods	CIFAR10		CIFAR100		TinyImageNet		ImageNet	
	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}
Joint Training‡ (Soft Upper Bound)	96.03		79.89		53.05		69.26	
EWC++ (Kirkpatrick et al., 2017)	57.34±2.10	60.33±2.73	35.35±1.96	38.78±2.32	22.26±1.15	24.39±1.18	24.81	26.21
BiC (Wu et al., 2019)	58.38±0.54	61.49±0.68	33.51±3.04	37.61±3.00	22.80±0.94	24.90±1.07	27.41	28.38
ER-MIR (Aljundi et al., 2019a)	57.28±2.43	61.93±3.35	35.35±1.41	38.28±1.15	22.10±1.14	24.54±1.26	20.48	20.68
GDumb (Prabhu et al., 2020)	53.20±1.93	55.27±2.69	32.84±0.45	34.03±0.89	18.17±0.19	18.69±0.45	14.41	14.21
RM† (Bang et al., 2021)	23.00±1.43	61.52±3.69	8.63±0.19	33.27±1.59	5.74±0.30	17.04±0.77	6.22	28.30
Baseline-ER (Sec. A.1)	57.46±2.25	60.17±2.96	35.61±2.08	39.10±2.02	22.45±1.15	24.54±1.26	25.16	26.50
CLIB (Ours)	**70.26±1.28**	**73.90±0.22**	**46.67±0.79**	**49.22±0.79**	**23.87±0.68**	**25.05±0.52**	**28.16**	**28.88**

Table 1: Comparison of online CL methods on the i-Blurry setup for CIFAR10, CIFAR100, TinyImageNet and ImageNet. 'Joint Training‡' shows the final accuracy of non-CL joint training as a soft upper bound where all relevant hyper-parameters were kept consistent with other compared CL methods. CLIB outperforms all other CL methods by large margins on both the A_{AUC} and the A_{avg}.

Results

i-Blurry-50-10 Results

- Memory size
  - CIFAR10: 500, CIFAR100: 2000, TinyImageNet: 4000, ImageNet: 20000
- # of Task
  - CIFAR10, CIFAR100, TinyImageNet: 5 tasks, ImageNet: 10 tasks

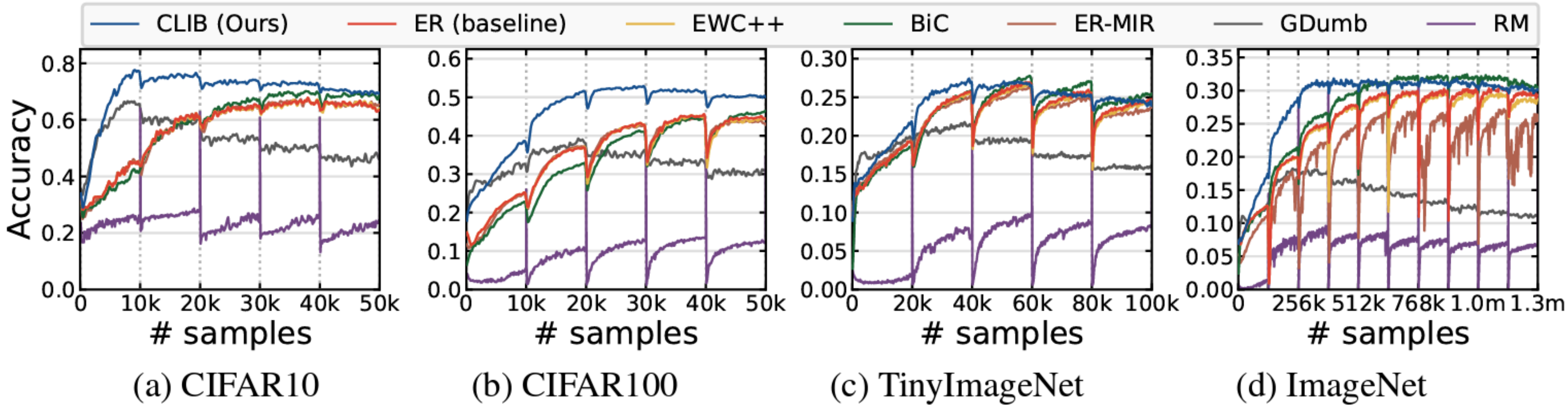


Figure 4: Accuracy-to-{number of samples} for various CL methods on CIFAR10, CIFAR100, TinyImageNet and ImageNet Our CLIB is consistent at maintaining high accuracy throughout inference while other CL methods are not as consistent.

Results
Various Values of N, M

CLIB: Continual Learning for i-Blurry task

Varying N	$N = 0$ (Blurry)		$N = 50$ (i-Blurry)		$N = 100$ (Disjoint)	
	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}
EWC++	53.24±0.56	57.04±0.94	57.34±2.10	60.33±2.73	77.64±1.81	77.80±1.93
BiC	51.51±0.12	54.88±0.30	58.38±0.54	61.49±0.68	**78.78±1.52**	**80.12±2.20**
ER-MIR	52.21±0.85	56.33±0.47	57.28±2.43	61.93±3.35	76.49±1.97	78.20±2.01
GDumb	45.86±0.80	46.37±2.09	53.20±1.93	55.27±2.69	65.27±1.54	66.74±2.39
RM†	22.54±1.11	54.07±0.70	23.00±1.43	61.52±3.69	33.17±3.71	66.67±2.38
Baseline-ER	53.28±0.57	57.13±1.01	57.46±2.25	60.17±2.96	77.82±2.06	77.47±2.69
CLIB (Ours)	**68.87±0.83**	**72.79±0.96**	**70.26±1.28**	**73.90±0.22**	78.58±2.09	77.96±3.28

Varying M	$M = 10$		$M = 30$		$M = 50$	
	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}
EWC++	57.34±2.10	60.33±2.73	65.71±2.20	69.94±3.01	68.01±0.85	73.26±2.33
BiC	58.38±0.54	61.49±0.68	65.88±3.24	70.31±4.88	68.08±3.72	73.33±1.21
ER-MIR	57.28±2.43	61.93±3.35	65.99±2.28	70.47±3.41	68.13±0.65	73.33±1.21
GDumb	53.20±1.93	55.27±2.69	54.73±1.54	54.63±2.39	53.86±0.59	52.82±1.24
RM†	23.00±1.43	61.52±3.69	26.60±1.74	61.52±0.48	28.32±5.08	59.32±4.57
Baseline-ER	57.46±2.25	60.17±2.96	65.92±2.25	70.04±2.87	68.26±0.84	73.16±1.49
CLIB (Ours)	**70.26±1.28**	**73.90±0.22**	**75.04±2.81**	**77.87±2.57**	**75.14±1.27**	**78.30±2.01**

Table 2: Analysis on various values of N (top) and M (bottom) in the i-Blurry-N-M setup using CIFAR10 dataset. For varying N, we use $M = 10$. For varying M, we use $N = 50$. Note that $N = 0$ corresponds to the blurry split and $N = 100$ corresponds to the disjoint split. For $N = 100$, CLIB outperforms or performs on par with other CL methods. For $N = 0$, the gap between CLIB and other methods widens. For $N = 50$, CLIB again outperforms all comparisons by large margins. For varying M, CLIB outperforms all comparisons excepting only the A_{avg} when $M = 0$.

Results
Ablation Study

Methods	CIFAR10		CIFAR100	
	A_{AUC}	A_{avg}	A_{AUC}	A_{avg}
CLIB	**70.26±1.28**	**73.90±0.22**	**46.67±0.79**	**49.22±0.79**
w/o Sample Importance Mem. (Sec.4.2)	53.75±2.11	56.31±2.56	36.59±1.22	38.59±1.21
w/o Memory-only training (Sec.4.3)	67.06±1.51	71.65±1.87	44.63±0.22	48.66±0.39
w/o Adaptive LR scheduling (Sec.4.4)	69.70±1.34	73.06±1.35	45.01±0.22	48.97±0.12

Table 3: Ablations for proposed components of our method using CIFAR10 and CIFAR100 dataset. All proposed components improve the performance, with sample-wise importance based memory management providing the biggest gains. While adaptive LR scheduling provides small gains in CIFAR10, the gains increase in the more challenging CIFAR100.

Conclusions

1. Proposing a new CL setup called i-Blurry, which addresses a more realistic setting that is **online, task-free, class-incremental, of blurry task boundaries**, and subject to **any-time inference**

2. Proposing a novel online and task-free CL method by a new **memory management**, **memory usage**, and **learning rate scheduling strategy**.

3. Outperforming existing CL models by large margins on multiple datasets and settings.

4. Proposing a **new metric** to better measure a CL model's capability for the desirable **any-time inference.**

Q & A

Thank you for Listening